National Institute of Technology Calicut
Department of Computer Science and Engineering
CS4053E – Data Mining | Winter 2025-26

Assignment 2 – Evaluation Report

Music Genre Classification (GTZAN Dataset)

Group Number	GROUP – 4
Dataset	GTZAN features_30_sec.csv (Kaggle)
Submission Deadline	5th April 2026, 10:00 PM
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1. Introduction

This report documents the experimental setup and evaluation results for Assignment 2 of CS4053E Data Mining. Two feature extraction techniques — Principal Component Analysis (PCA) and ANOVA F-test SelectKBest — are applied alongside three classification algorithms (Decision Tree, K-Nearest Neighbours, Gaussian Naïve Bayes) to the GTZAN Music Genre Dataset. Hyperparameters are tuned using GridSearchCV with 5-fold Stratified Cross-Validation. All models are evaluated on a held-out 20% test set using Accuracy, Precision (macro), Recall (macro), and F1-Score (macro).

2. Dataset Overview

The GTZAN Music Genre Dataset (features_30_sec.csv) contains pre-extracted numerical audio features for 1,000 tracks across 10 balanced genres (100 tracks each). After dropping non-informative columns (filename, length), 57 numerical features remain.

Property	Value
Total Instances	1,000 songs
Number of Genres	10 (Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae, Rock)
Tracks per Genre	100 (balanced)
Numerical Features Used	57 (after removing filename, length, label)
Train / Test Split	80% / 20% — 800 train | 200 test (stratified, random_state=42)
Preprocessing Applied	Z-score Standardisation via StandardScaler (fit on train only)
Cross-Validation	StratifiedKFold, 5 folds, shuffled, random_state=42

Hyperparameter Tuning	GridSearchCV (scoring = accuracy, n_jobs = -1)

Table 1: Dataset Properties

3. Feature Extraction Methods

3.1 Method 1 – Principal Component Analysis (PCA)

PCA transforms the 57 correlated audio features into uncorrelated principal components ordered by explained variance. A 90% variance-retention threshold is used: **PCA(n_components=0.90)** automatically selected **23 components**, capturing **90.46% of the total variance** and reducing the feature space by 60% (from 57 to 23 dimensions). The scree plot below confirms a rapid variance drop-off after the first few components.

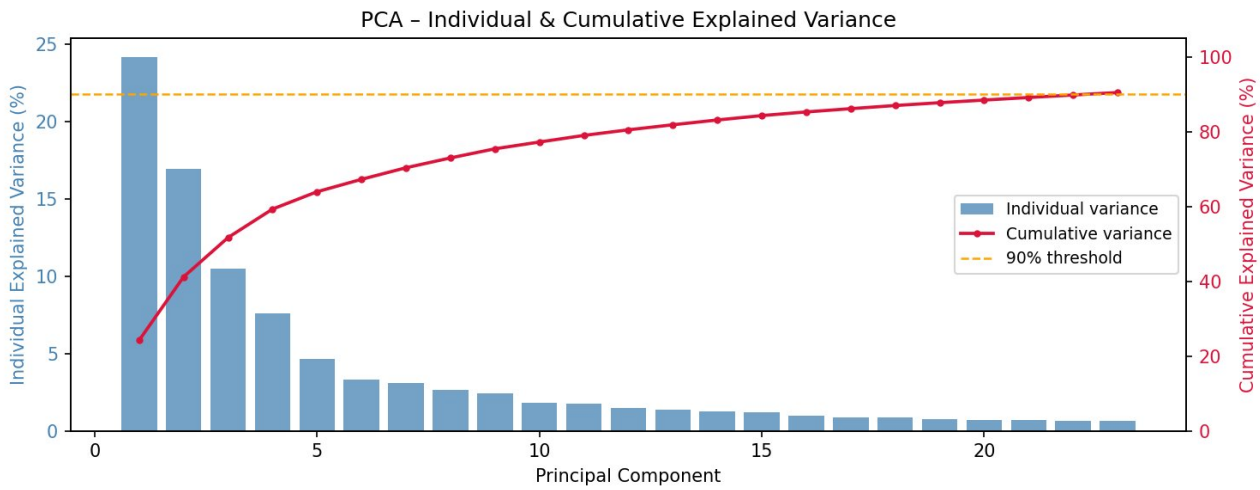

Figure 1: PCA Scree Plot — Individual & Cumulative Explained Variance

3.2 Method 2 – SelectKBest (ANOVA F-test)

SelectKBest with the ANOVA F-test ranks each feature by its ability to discriminate between the 10 genre classes, preserving the original interpretable audio features. The number of features k is treated as a hyperparameter and tuned over {10, 20, 30, 40} via GridSearchCV independently for each classifier. The chart below visualises the **top 20 features by ANOVA F-score** for reference; chroma_stft_mean dominates at F ≈ 142. The best k found by grid search was **k=30 for Decision Tree** and **k=40 for both KNN and Naïve Bayes**.

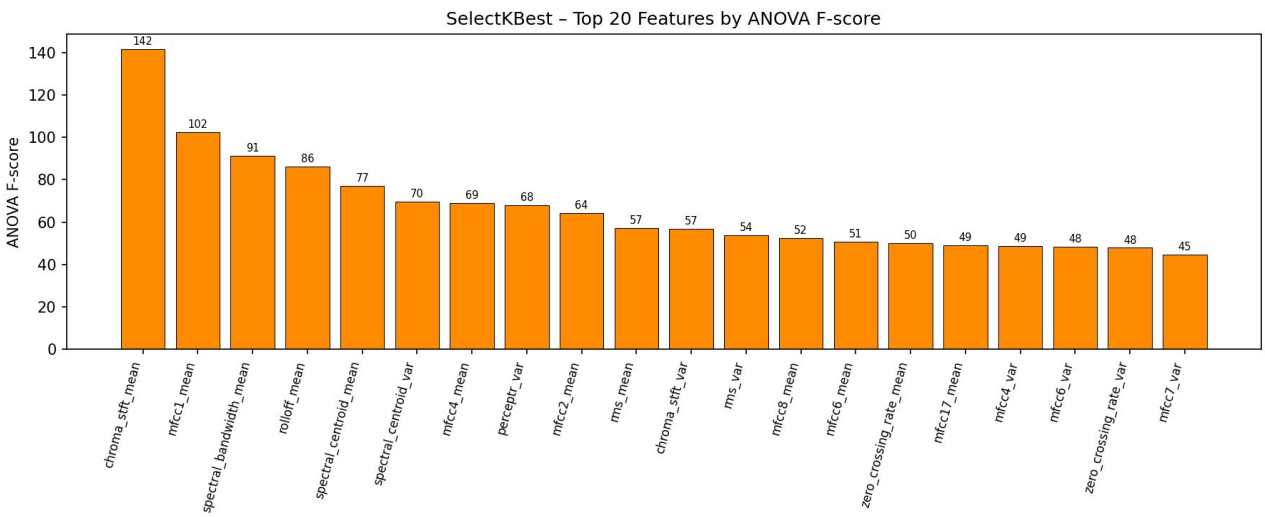

Figure 2: SelectKBest — Top 20 Features by ANOVA F-score (best k: Decision Tree=30, KNN=40, Naïve Bayes=40)

Property	PCA	SelectKBest (ANOVA F-test)
Input features	57 (standardised)	57 (standardised)
Output features	23 components	40 features (best k = 40 for KNN/NB; k = 30 for Decision Tree)
Variance captured	90.46%	N/A (filter-based method)
Interpretability	Low (abstract latent axes)	High (original audio features)
Captures correlation	Yes (orthogonal PCs)	No (univariate ranking)
Reduction ratio	60% (57 → 23)	30% (57 → 40) / 47% (57 → 30)

Table 2: Comparison of Feature Extraction Methods

4. Classification Models and Grid Search

Three scikit-learn classifiers are trained and tuned using GridSearchCV with 5-fold Stratified Cross-Validation. The best estimator from each search is evaluated on the 20% held-out test set.

Classifier	Hyperparameter	Values Searched
Decision Tree	criterion	gini, entropy
Decision Tree	max_depth	5, 10, 15, 20, None
Decision Tree	min_samples_split	2, 5, 10
Decision Tree	min_samples_leaf	1, 2, 4
KNN	n_neighbors	3, 5, 7, 9, 11
KNN	weights	uniform, distance
KNN	metric	euclidean, manhattan
Naive Bayes	var_smoothing	1e-11, 1e-10, 1e-9, 1e-8, 1e-7, 1e-6
SelectKBest (k)	k	10, 20, 30, 40 (tuned per classifier)

Table 3: Grid Search Hyperparameter Space

5. Experimental Results

Train set: 800 samples (80%) | Test set: 200 samples (20%) | All Precision, Recall and F1 values are macro-averaged across 10 genre classes. CV Accuracy is the best GridSearchCV cross-validation score on the training set. Best-performing configuration (KNN + SelectKBest) is highlighted in green.

5.1 Feature Extraction Method 1: PCA (23 components, 90.46% variance retained)

Classifier	Best Parameters	CV Acc.	Test Acc.	Precision	Recall	F1
Decision Tree	criterion=entropy max_depth=10 min_samples_leaf=1 min_samples_split=5	49.25%	50.00%	50.54%	50.00%	50.06%

KNN	metric=manhattan n_neighbors=5 weights=distance	67.12%	70.50%	71.04%	70.50%	69.94%
Naive Bayes	var_smoothing=1e-11	57.00%	56.00%	55.72%	56.00%	55.39%

Table 4: PCA Feature Extraction — Results

5.2 Feature Extraction Method 2: SelectKBest ANOVA F-test (k tuned per classifier)

Classifier	Best Parameters	CV Acc.	Test Acc.	Precision	Recall	F1
Decision Tree	criterion=entropy, k=30 max_depth=15 min_samples_leaf=1 min_samples_split=2	51.88%	52.50%	53.81%	52.50%	52.61%
KNN	k=40, metric=manhattan n_neighbors=5 weights=distance	66.75%	71.00%	71.90%	71.00%	71.12%
Naive Bayes	k=40 var_smoothing=1e-11	53.50%	56.50%	56.85%	56.50%	54.77%

Table 5: SelectKBest Feature Extraction — Results

5.3 Combined Performance Summary — All 6 Experiments

Feature Method	Classifier	CV Acc.	Test Acc.	Precision (Macro)	Recall (Macro)	F1 (Macro)
PCA	Decision Tree	49.25%	50.00%	50.54%	50.00%	50.06%
PCA	KNN	67.12%	70.50%	71.04%	70.50%	69.94%
PCA	Naive Bayes	57.00%	56.00%	55.72%	56.00%	55.39%
SelectKBest	Decision Tree	51.88%	52.50%	53.81%	52.50%	52.61%
SelectKBest	**KNN**	**66.75%**	**71.00%**	**71.90%**	**71.00%**	**71.12%**
SelectKBest	Naive Bayes	53.50%	56.50%	56.85%	56.50%	54.77%

Table 6: Combined Performance Summary — best model (KNN + SelectKBest) highlighted in green

5.4 Visualisations

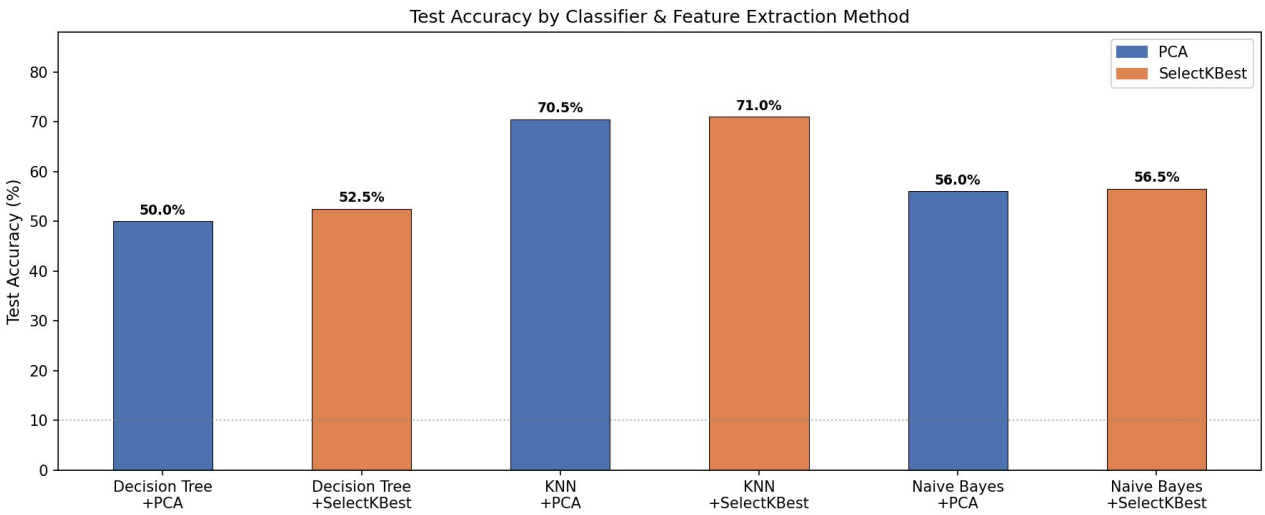

Figure 3: Test Accuracy by Classifier & Feature Extraction Method

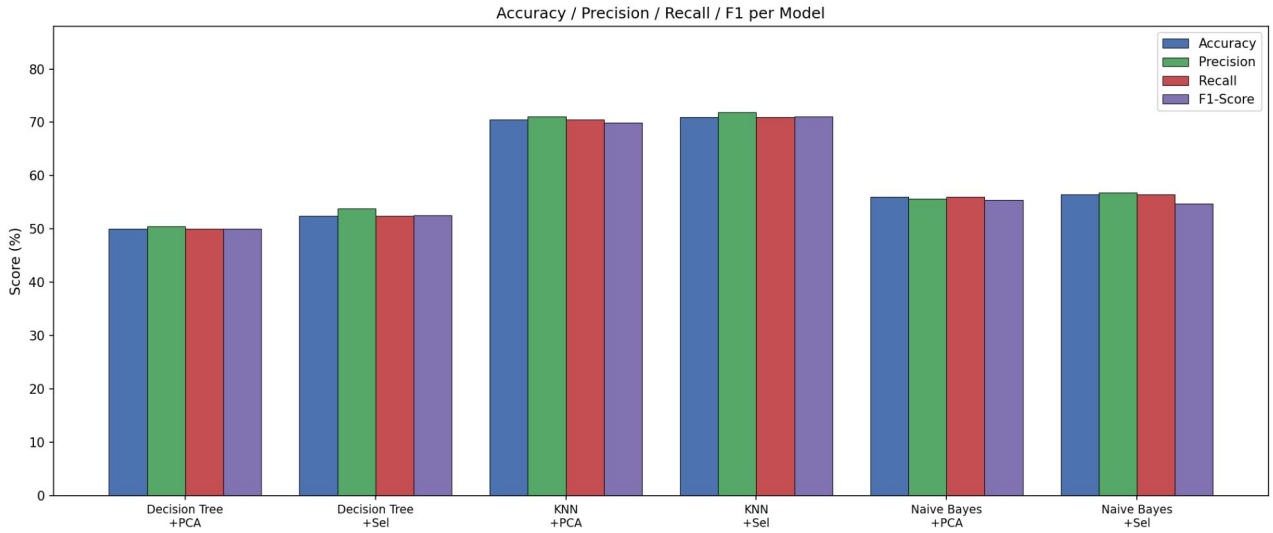

Figure 4: Accuracy / Precision / Recall / F1 per Model (macro-averaged)

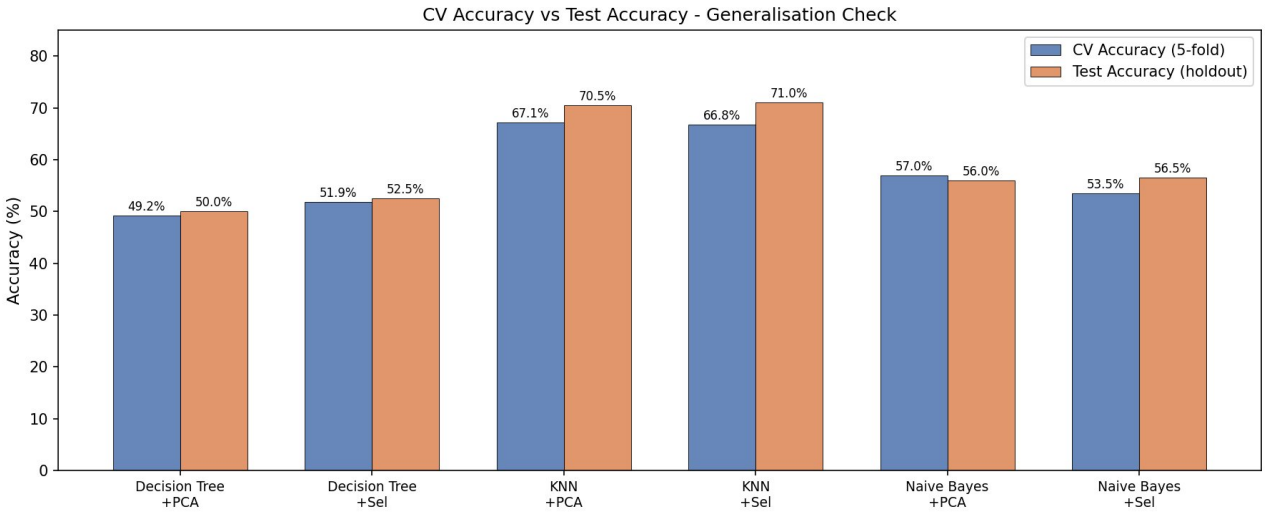

Figure 5: CV Accuracy vs Test Accuracy — Generalisation Check

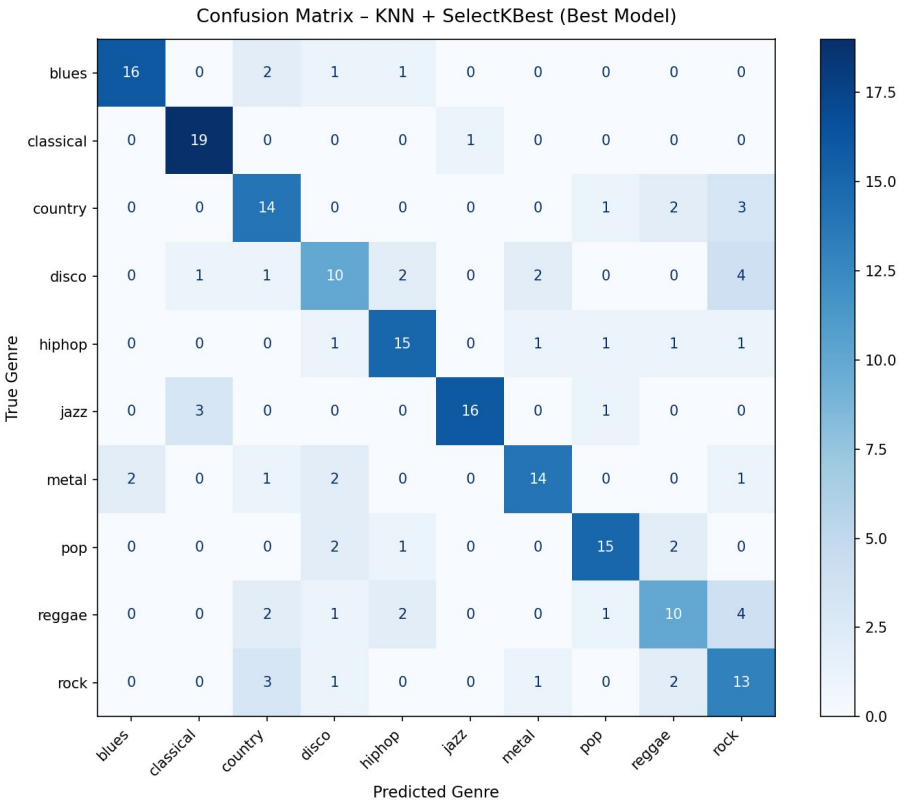

Figure 6: Confusion Matrix — KNN + SelectKBest (Best Model)

6. Analysis and Discussion

6.1 Impact of Feature Extraction Method

Both methods produce competitive results. SelectKBest marginally outperforms PCA for KNN (71.0% vs 70.5%) and Decision Tree (52.5% vs 50.0%), while PCA holds a tiny edge for Naive Bayes (56.0% vs 56.5%). PCA produces **23 orthogonal components** capturing 90.46% of total variance, encoding inter-feature correlations holistically. SelectKBest independently ranks all 57 features by ANOVA F-score; the optimal k was **30 for Decision Tree** and **40 for KNN and Naïve Bayes**, preserving original audio descriptors at the cost of ignoring feature correlations.

6.2 Classifier Analysis

Decision Tree: Worst performer — 50.0% (PCA) and 52.5% (SelectKBest, k=30). Even with entropy criterion and GridSearchCV-tuned depth (max_depth=10/15), axis-aligned splits cannot capture the smooth decision boundaries needed for continuous audio features.

K-Nearest Neighbours (KNN): Best classifier in both feature spaces — 70.5% with PCA and **71.0% with SelectKBest**. Manhattan distance with distance-weighted voting (n_neighbors=5, weights=distance) effectively captures local similarity structure in the audio feature space. This is the recommended model.

Naive Bayes (Gaussian): Mid-range at 56.0% (PCA) and 56.5% (SelectKBest, k=40). The class-conditional independence assumption limits performance on correlated audio features, but it is the fastest classifier and a useful baseline.

6.3 Generalisation Analysis

Figure 5 compares 5-fold CV accuracy to test accuracy for all 6 configurations. All models show test accuracy at or above CV accuracy, confirming no overfitting. KNN+PCA: CV=67.12% → Test=70.50% (+3.38 pp). KNN+SelectKBest: CV=66.75% → Test=71.00% (+4.25 pp). This positive gap is typical when

the held-out test set is slightly easier than the average CV fold for a well-regularised model. Decision Tree results are tightly bounded (CV≈Test), suggesting the depth constraints prevent overfitting but also limit expressive power.

6.4 Confusion Matrix and Per-Genre Analysis (Best Model: KNN + SelectKBest)

Figure 6 shows the confusion matrix for the best model. Classical music achieves the highest per-genre accuracy (19/20 = 95%), benefiting from highly distinctive tonal and harmonic features. Jazz (16/20 = 80%) and Blues (16/20 = 80%) also perform well. Disco and Reggae are the hardest to classify (both 10/20 = 50%): Disco is confused with Rock (4 errors) and Hip-Hop (2 errors); Reggae is confused with Rock (4 errors) and Country (2 errors). These errors are musically intuitive — reggae and rock share rhythmic patterns, and disco shares tempo with hip-hop.

Genre	Correct/Total	Accuracy	Precision	Recall	F1-Score
Blues	16 / 20	80.0%	0.89	0.80	0.84
Classical	19 / 20	95.0%	0.83	0.95	0.88
Country	14 / 20	70.0%	0.61	0.70	0.65
Disco	10 / 20	50.0%	0.56	0.50	0.53
Hip-Hop	15 / 20	75.0%	0.71	0.75	0.73
Jazz	16 / 20	80.0%	0.94	0.80	0.86
Metal	14 / 20	70.0%	0.78	0.70	0.74
Pop	15 / 20	75.0%	0.79	0.75	0.77
Reggae	10 / 20	50.0%	0.59	0.50	0.54
Rock	13 / 20	65.0%	0.50	0.65	0.57
Macro Avg	142 / 200	71.0%	0.72	0.71	0.71

Table 7: Per-Genre Classification Report — KNN + SelectKBest (Best Model)

7. Best Model Configuration

Aspect	Best Configuration
Feature Extraction Method	SelectKBest (ANOVA F-test)
Best k (features selected)	40
Classifier	K-Nearest Neighbours (KNN)
Best Parameters	n_neighbors=5, metric=manhattan, weights=distance
CV Accuracy (5-fold)	66.75%
Test Accuracy	71.00%
Precision (Macro-avg)	71.90%
Recall (Macro-avg)	71.00%
F1-Score (Macro-avg)	71.12%

Table 8: Best Model Configuration

8. Conclusion

Among all six configurations, **KNN with SelectKBest (k=40)** achieves the highest test accuracy of **71.0%** (Precision 71.9%, Recall 71.0%, F1 71.1%) on the GTZAN 10-class music genre task. KNN with PCA (23 components, 90.46% variance) is a close second at 70.5%, confirming that KNN is the most effective classifier for this dataset regardless of the feature extraction method.

PCA compresses 57 features to **23 orthogonal components** retaining 90.46% variance. SelectKBest retains the **top 40 features** (k=40 for KNN/NB, k=30 for Decision Tree) by ANOVA F-score, preserving interpretability: chroma_stft_mean, mfcc1_mean, and spectral_bandwidth_mean are the most discriminative audio descriptors.

Decision Tree performs worst (50–52.5%) across all conditions and is not recommended for this dataset without ensemble methods (e.g., Random Forests). Naive Bayes is computationally efficient but limited by its independence assumption. Future work could explore SVM or gradient-boosted ensembles to push accuracy beyond 71%.